Skill – 2

```
jamuna@Jem:~$ sudo ip netns del host1
jamuna@Jem:~$ sudo ip netns del router
jamuna@Jem:~$ sudo ip netns del host2
jamuna@Jem:~$ sudo ip netns add host1
jamuna@Jem:~$ sudo ip netns add router
jamuna@Jem:~$ sudo ip netns add host2
jamuna@Jem:~$ sudo ip link add veth-h1 type veth peer name veth-r1
jamuna@Jem:~$ sudo ip link add veth-r2 type veth peer name veth-h2
jamuna@Jem:~$ sudo ip link set veth-h1 netns host1
jamuna@Jem:~$ sudo ip link set veth-r1 netns router
jamuna@Jem:~$ sudo ip link set veth-r2 netns router
jamuna@Jem:~$ sudo ip link set veth-h2 netns host2
jamuna@Jem:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
jamuna@Jem:~$ sudo ip netns exec host1 ip link set veth-h1 up
jamuna@Jem:~$ sudo ip netns exec host1 ip link set lo up
jamuna@Jem:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
jamuna@Jem:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
jamuna@Jem:~$ sudo ip netns exec router ip link set veth-r1 up
jamuna@Jem:~$ sudo ip netns exec router ip link set veth-r2 up
jamuna@Jem:~$ sudo ip netns exec router ip link set lo up
jamuna@Jem:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
jamuna@Jem:~$ sudo ip netns exec host2 ip link set veth-h2 up
jamuna@Jem:~$ ^C
jamuna@Jem:~$ sudo ip netns exec host2 ip link set lo up
jamuna@Jem:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
jamuna@Jem:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
jamuna@Jem:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
```

```
jamuna@Jem:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
jamuna@Jem:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
jamuna@Jem:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.089 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.066 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.081 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2051ms
rtt min/avg/max/mdev = 0.066/0.078/0.089/0.009 ms
```

```
jamuna@Jem:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
jamuna@Jem:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 13 limit 1000 delay 20ms loss 1% seed 6578717596086239840
jamuna@Jem:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
jamuna@Jem:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.0K Nov 11 03:11 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.0K Nov 11 03:11 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.2K Nov 11 03:11 /tmp/cap3.pcap
jamuna@Jem:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      5 64
jamuna@Jem:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      5 64
jamuna@Jem:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      4 64
```

Terminal 1 -

```
jamuna@Jem:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
17 ^C
jamuna@Jem:~$
```

Terminal 2 -

```
jamuna@Jem:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
17 ^C
jamuna@Jem:~$
```

Terminal 3 -

```
jamuna@Jem:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
19 ^C
jamuna@Jem:~$
```

Terminal 4 -

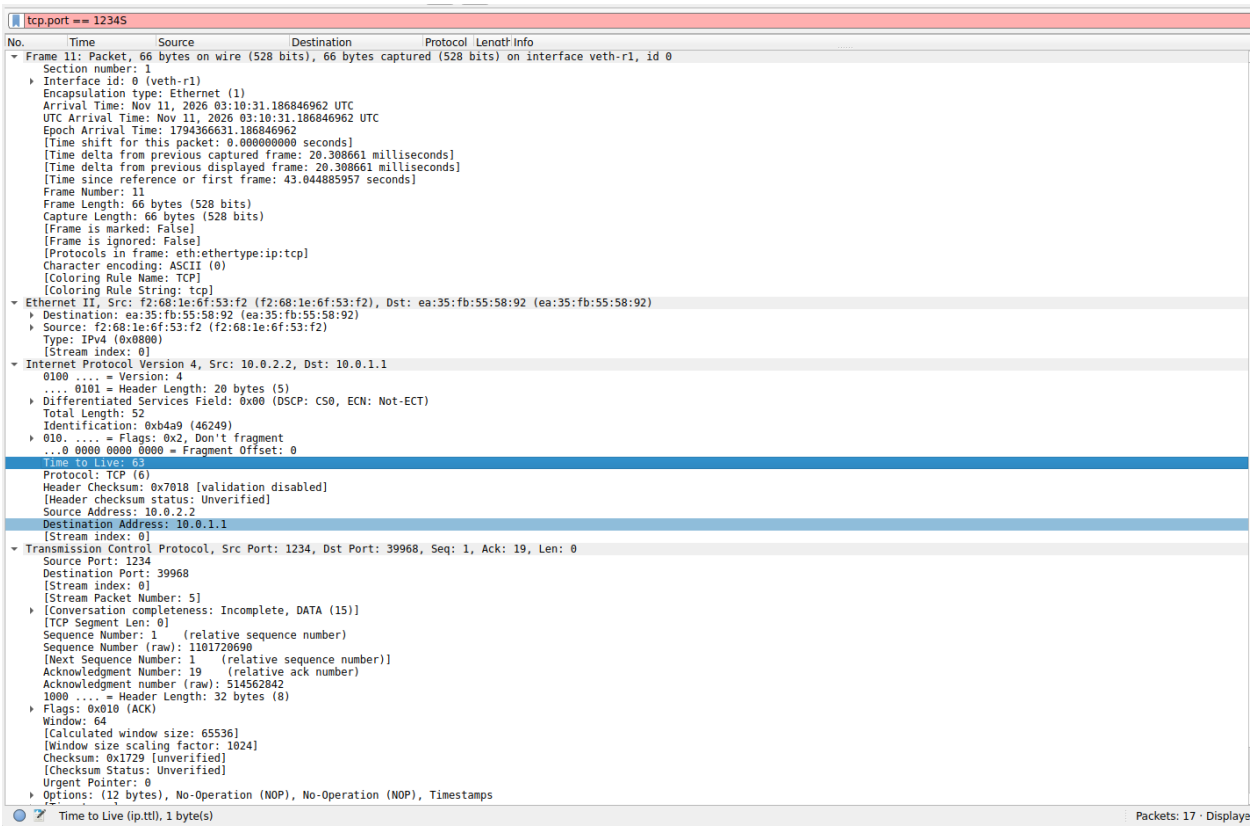
```
jamuna@Jem:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host1!
Computer Networks Lab
Three Point Capture Experiment
^C
jamuna@Jem:~$
```

Terminal 5 -

```
jamuna@Jem:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
Computer Networks Lab
Three Point Capture Experiment
^C
jamuna@Jem:~$
```

```
jamuna@Jem:~$ wireshark /tmp/cap2.pcap &
[1] 11546
jamuna@Jem:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a34 3336 6636 6437  431210a..436f6d7
00000030: 3037 3537 3436 3537 3232 3034 6536 3537  075746572204e657
00000040: 3437 3736 6637 3236 6237 3332 3034 6336  4776f726b73204c6
00000050: 3136 3230 610a 0a35 3436 3837 3236 3536  1620a..546872656
00000060: 3532 3035 3036 6636 3936 6537 3432 3034  520506f696e74204
00000070: 3336 3137 3037 3437 3537 3236 3532 3034  3617074757265204
00000080: 3537 3837 3036 3537 3236 3936 6436 3536  578706572696d656
00000090: 6537 3430 610a 0a                        e740a..
jamuna@Jem:~$
```




tcp.port == 1234S
No.
Time
Source
Destination
Protocol
Lenath
Info
Frame 11: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface veth-r1, id 0
Section number: 1
Interface id: 0 (veth-r1)
Encapsulation type: Ethernet (1)
Arrival Time: Nov 11, 2026 03:10:31.186846962 UTC
UTC Arrival Time: Nov 11, 2026 03:10:31.186846962 UTC
Epoch Arrival Time: 1794366631.186846962
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 20.308661 milliseconds]
[Time delta from previous displayed frame: 20.308661 milliseconds]
[Time since reference or first frame: 43.044885957 seconds]
Frame Number: 11
Frame Length: 66 bytes (528 bits)
Capture Length: 66 bytes (528 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: f2:68:1e:6f:53:f2 (f2:68:1e:6f:53:f2), Dst: ea:35:fb:55:58:92 (ea:35:fb:55:58:92)
Destination: ea:35:fb:55:58:92 (ea:35:fb:55:58:92)
Source: f2:68:1e:6f:53:f2 (f2:68:1e:6f:53:f2)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.0.2.2, Dst: 10.0.1.1
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xb4a9 (46249)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 63
Protocol: TCP (6)
Header Checksum: 0x7018 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.2
Destination Address: 10.0.1.1
[Stream index: 0]
Transmission Control Protocol, Src Port: 1234, Dst Port: 39968, Seq: 1, Ack: 19, Len: 0
Source Port: 1234
Destination Port: 39968
[Stream index: 0]
[Stream Packet Number: 5]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1101720690
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 19 (relative ack number)
Acknowledgment number (raw): 514562842
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 64
[Calculated window size: 65536]
[Window size scaling factor: 1024]
Checksum: 0x1729 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
Time to Live (ip.ttl), 1 byte(s)
Packets: 17 · Displaye